NUMBER SYSTEM MADE EASY

VIDEO 10 • THE FINAL CONVERSION

Octal ↔️ Hexadecimal

Use binary as the bridge between the two systems.

OCTAL → BINARY → HEXADECIMAL

Learning focus

By the end of this lesson, you will be able to convert in both directions.

OCTAL → HEXADECIMAL

Convert octal digits to 3-bit binary groups, then regroup into 4-bit groups.

HEXADECIMAL → OCTAL

Convert hexadecimal digits to 4-bit binary groups, then regroup into 3-bit groups.

Binary acts as the bridge.

The conversion pathway

Octal and hexadecimal do not have a direct grouping relationship.

DIRECTION 1

OCTAL → BINARY → HEXADECIMAL

DIRECTION 2

HEXADECIMAL → BINARY → OCTAL

Why binary? Each octal digit = 3 bits • Each hexadecimal digit = 4 bits

Remember the bit groups

These are the rules that make the conversion work.

OCTAL → BINARY

Replace each octal digit with exactly 3 binary bits.

Example: 5 → 101

HEXADECIMAL → BINARY

Replace each hexadecimal digit with exactly 4 binary bits.

Example: F → 1111

From binary to octal: group 3 • From binary to hexadecimal: group 4

Worked example 1: Octal → Hexadecimal

Convert 57_8 to hexadecimal.

STEP 1 • OCTAL → BINARY

5 → 101
7 → 111

$57_8 = 101111_2$

→

STEP 2 • GROUP 4

101111

Pad on the left:
0010 | 1111

→

STEP 3 • HEX

0010 → 2
1111 → F

$2F_{16}$

$$57_8 = 2F_{16}$$

Worked example 2: Hexadecimal → Octal

Convert $2F_{16}$ to octal.

STEP 1 • HEX → BINARY

2 → 0010
F → 1111

00101111_2

→

STEP 2 • GROUP 3

00 | 101 | 111

Pad on the left:
000 | 101 | 111

→

STEP 3 • OCTAL

000 → 0
101 → 5
111 → 7

Remove leading 0

$$2F_{16} = 57_8$$

Important reminder

Follow the correct pathway and group from the right.

OCTAL → HEXADECIMAL

Octal → Binary → Hexadecimal

3 bits per octal digit
Then group into 4 bits.

HEXADECIMAL → OCTAL

Hexadecimal → Binary → Octal

4 bits per hexadecimal digit
Then group into 3 bits.

If a group is short, add zeros to the LEFT.

Practice question

Try it yourself before revealing the answer.

Convert $17_8 \rightarrow ?_{16}$

You have 20 seconds!

STEP 1

1 → 001 7 → 111
Binary: 001111

STEP 2

Group into 4: 0011 | 11
Pad: 0011 | 0011

Answer reveal on next slide →

Practice answer

Both 4-bit groups represent the hexadecimal digit 3.

0011 | 0011

0011 → 3 **0011 → 3**

CORRECT ANSWER

$17_8 = 33_{16}$

Final challenge

Solve independently before looking at the answer.

Convert $3C_{16} \rightarrow ?_{8}$

You have 30 seconds!

CLUE 1 • HEX → BINARY

3 → 0011
C → 1100
Binary: 00111100

CLUE 2 • GROUP 3

00 | 111 | 100
Pad the left group with a zero.

Pause here and calculate the octal digits.

Final challenge: answer reveal

Convert each 3-bit group into one octal digit.

000 | 111 | 100

000 → 0 **111 → 7** **100 → 4**

REMOVE THE LEADING ZERO

$3C_{16} = 074_8 = 74_8$

Congratulations if you got it!

The complete 10-video journey

You have now reached the final conversion in the series.

1 Introduction to Number Systems

2 Binary ↔ Octal

3 Binary → Decimal

4 Decimal → Binary

5 Binary ↔ Hexadecimal

6 Decimal → Octal

7 Octal → Decimal

8 Decimal → Hexadecimal

9 Hexadecimal → Decimal

10 Octal ↔ Hexadecimal

From understanding number systems to converting between them!

Conversion methods: final recap

Choose the method that matches the direction of conversion.

GROUPING

Binary ↔ Octal
3 bits

Binary ↔ Hex
4 bits

POSITIONAL

Binary → Decimal
Octal → Decimal
Hexadecimal → Decimal

DIVISION-REMAINDER

Decimal → Binary
Decimal → Octal
Decimal → Hexadecimal

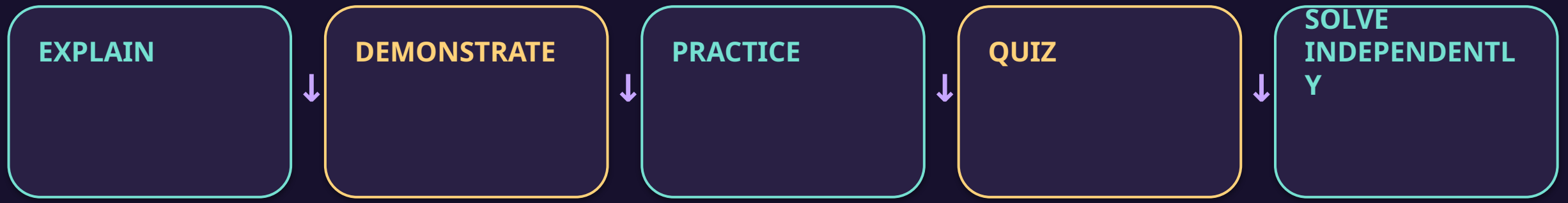
BINARY BRIDGE

Octal ↔ Hexadecimal

For octal and hexadecimal, use binary as the bridge.

Final message

Understand the process—not just the answers.



Practice makes conversion easier!

THANK YOU FOR LEARNING WITH NUMBER SYSTEM MADE EASY